

THE ONTOLOGICAL PROTOCOL

Canonical Edition (CE)

Structural Relational Monism under Deterministic Implementation

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Abstract

The Ontological Protocol establishes a strictly processual economic system grounded in Structural Relational Monism.

Value is not an object, not a token, and not a stored quantity. It is the measurable persistence of causal structure within a temporally ordered graph.

This Canonical Edition consolidates previous iterations into a fully deterministic, simulation-ready architecture. All recursive dependencies are temporally separated. All global quantities are bounded. All metrics are horizon-limited. No exogenous valuation enters the system.

The protocol separates:

- The **Ledger Layer** (irreversible thermodynamic record),
- The **Ripple Layer** (causal mass propagation),
- The **Density Layer** (Dynamic Causal Density),
- The **Regulation Layer** (cybernetic homeostasis).

No object ontology is reintroduced at any level. The economy is defined as a living, bounded, deterministic causal graph.

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1 Foundations

1.1 Structural Relational Monism

Axiom 1 (Structural Monism). There are no objects. There are only processes of varying temporal persistence.

Matter is a process with high persistence. Action is a process with high velocity. Code is a process in frozen potential.

No intrinsic value exists in states. Value exists only in structured transition.

Axiom 2 (Esse est Operari). To be is to execute. A node possesses ontological weight only insofar as it participates in causal interaction.

An isolated node has zero structural weight.

2 Formal Preliminaries

2.1 Time

Time is discrete.

$$T \in \mathbb{N}$$

Each epoch T represents a globally ordered computation interval.

No continuous time appears in the operational core.

2.2 Event Graph

Definition 1 (Event). An event e is defined as the tuple:

$$e = (\text{ID}, T(e), \text{Sig}(e), \text{Parents}(e), b(e))$$

where:

- ID is a unique identifier,
- $T(e)$ is the epoch index,
- $\text{Sig}(e)$ is a non-empty signer set,
- $\text{Parents}(e)$ is a finite set of prior events,
- $b(e) \geq 0$ is irreversible burn.

2.3 Deterministic Ordering

Events are totally ordered lexicographically by:

$$(T(e), \text{ID}(e))$$

All nodes compute identical event sequences.

2.4 Acyclicity

$$\forall p \in \text{Parents}(e) : T(p) < T(e)$$

Thus the event graph is a Directed Acyclic Graph (DAG).

3 Ledger Layer

3.1 Burn

Definition 2 (Burn). Burn $b(e)$ is permanently destroyed value sent to the null sink.

Burn is:

- irreversible,
- non-transferable,
- non-redistributable.

It serves purely as thermodynamic differentiation between signal and noise.

3.2 Thermodynamic Floor

Define the production floor for epoch T :

$$B_{prod}(T) = Q_{0.05}(\{b(e) \mid e \in \mathcal{E}_{T-1}\})$$

where $Q_{0.05}$ is the 5th percentile.

This adapts minimal burn cost to network activity.

3.3 Ledger Invariance

The ledger stores only:

- Event structure,
- Parent relations,
- Burn.

It does not store:

- Ripple values,
- Density metrics,
- Attribution weights,
- Income balances.

All value computation is derived, not recorded.

Proposition 1 (Ledger Neutrality). The ledger is ontologically neutral. All evaluative layers are externally computable from the event DAG.

4 Horizon Constraint

To prevent unbounded memory and structural oligarchy, define rolling horizon H :

$$H \in \mathbb{N}$$

Only events in epochs:

$$[T - H + 1, \dots, T]$$

participate in value computation.

Events older than H epochs persist in the ledger but do not contribute to Ripple or Density.

Remark 1. History exists, but structural power decays.

5 The Ripple Layer

5.1 Purpose

The Ripple Layer assigns **causal mass** to events.

Ripple is not stored value. It is a derived scalar field over the event DAG.

It measures the structural influence of an event within the rolling horizon.

5.2 Temporal Separation Principle

To prevent algebraic recursion and infinite feedback loops:

Axiom 3 (Temporal Separation). Attribution weights used in epoch T are derived solely from density computed in epoch $T - 1$.

Thus:

$$\alpha_T(e) = f(DCD_{T-1}(e))$$

No circular dependency exists within a single epoch.

5.3 Attribution Weights

Let $\mathcal{E}_{T-1}$ be all events in epoch $T - 1$.

Define total density:

$$D_{tot}(T - 1) = \sum_{u \in \mathcal{E}_{T-1}} DCD_{T-1}(u)$$

Attribution weight:

$$\alpha_T(e) = \begin{cases} \frac{DCD_{T-1}(e)}{D_{tot}(T - 1)} & \text{if } D_{tot}(T - 1) > 0, \\ 0 & \text{otherwise.} \end{cases}$$

To prevent structural capture:

$$0 \leq \alpha_T(e) \leq \alpha_{\max} < 1$$

where $\alpha_{\max}$ is a protocol constant.

5.4 Active Persistence

Let:

- $\text{Depth}_H(e)$ = maximum path length to e within horizon H ,
- $\Delta T(e)$ = epochs since last structural interaction.

Define active persistence:

$$\lambda_T(e) = \left(1 - \frac{1}{1 + \text{Depth}_H(e)}\right) \cdot e^{-\frac{\Delta T(e)}{\tau}}$$

where $\tau > 0$ is decay constant.

Properties:

- Depth increases persistence,

- Inactivity exponentially decreases it,
- $\lambda_T(e) \in [0, 1)$.

Remark 2. No node remains structurally immortal.

5.5 Ripple Recursion

Define contraction factor:

$$0 \leq \gamma < 1$$

Let $Children(e)$ denote all events referencing e within horizon H .

Define Ripple recursively:

$$R_T(e) = b(e) + \gamma \sum_{y \in Children(e)} \frac{1}{|Parents(y)|} \cdot R_T(y) \cdot \alpha_T(y) \cdot \lambda_T(e)$$

Interpretation:

- Direct burn anchors structural mass.
- Downstream influence propagates backward.
- Propagation is weighted by:
 - attribution α_T ,
 - active persistence λ_T ,
 - contraction γ .

5.6 Existence and Uniqueness

Proposition 2 (Convergence). Given:

- Finite horizon H ,
- $\gamma < 1$,
- Bounded $\alpha_T(e)$,
- Finite event set per epoch,

the Ripple system admits a unique finite solution.

Sketch. The recursion defines a contraction mapping over a finite-dimensional vector space. Since $\gamma\alpha_{\max} < 1$, spectral radius is bounded below 1. Thus convergence follows from Banach fixed-point theorem. $\square$

5.7 No Infinite Amplification

Proposition 3. Ripple mass is bounded above by:

$$R_T(e) \leq \frac{b_{\max}}{1 - \gamma\alpha_{\max}}$$

Thus no event can accumulate infinite structural mass.

5.8 Interpretation

Ripple measures:

- Not ownership,
- Not stored wealth,
- Not price.

It measures causal embeddedness within the living graph.

Remark 3. Value is not held. Value is structurally sustained.

6 Dynamic Causal Density

6.1 Purpose

Dynamic Causal Density (DCD) measures the structured propagation of causal influence within the rolling horizon.

Ripple measures causal mass. Density measures structured expansion.

6.2 Rolling Horizon Domain

All density computation is restricted to:

$$\mathcal{H}_T = \{e \mid T - H + 1 \leq T(e) \leq T\}$$

Events outside $\mathcal{H}_T$ contribute neither to Ripple nor to Density.

6.3 Effective Branching

Let:

- $Children(e)$ = referencing events in $\mathcal{H}_T$
- $U_1(e)$ = number of distinct first-degree neighbors
- $U_2(e)$ = number of distinct second-degree neighbors

To prevent clique inflation, define capped sampling:

$$\tilde{U}_1(e) = \min(U_1(e), K_1)$$

$$\tilde{U}_2(e) = \min(U_2(e), K_2)$$

where K_1, K_2 are protocol constants.

6.4 Local Diversity Index

Define:

$$\Omega_T(e) = \frac{\log(1 + \tilde{U}_2(e))}{\log(1 + \tilde{U}_1(e)^2)}$$

Properties:

- $\Omega_T(e) \in [0, 1]$

- Clique-like structures collapse toward 0
- Bridge-like propagation approaches 1

Remark 4. Diversity measures expansion beyond closed clusters.

6.5 Effective Structural Propagation

Define structural expansion factor:

$$\sigma_T(e) = \tilde{U}_1(e) \cdot \Omega_T(e)$$

Thus:

- Pure repetition yields low Ω
- Pure isolation yields low $\tilde{U}_1$
- Organic propagation maximizes both

6.6 Density Definition

Define:

$$DCD_T(e) = \sum_{k=T-H+1}^T \sigma_k(e) \cdot \lambda_k(e) \cdot R_k(e)$$

where:

- $\sigma_k(e)$ = structural expansion at epoch k
- $\lambda_k(e)$ = active persistence
- $R_k(e)$ = Ripple mass

6.7 Boundedness

Proposition 4. Given bounded $\sigma_k(e)$, bounded $\lambda_k(e)$, and bounded $R_k(e)$, the density $DCD_T(e)$ is finite.

Sketch. All components are bounded in $[0, C]$. The sum spans finite horizon H . Thus $DCD_T(e) \leq H \cdot C^3$. $\square$

6.8 Sybil Resistance

Proposition 5. If a node generates a clique-like subgraph with no novel second-degree propagation, then $\Omega_T(e) \rightarrow 0$ and thus $DCD_T(e) \rightarrow 0$.

Remark 5. Sybil clusters generate structure without diversity. Structure without diversity carries no density.

6.9 No Stored Reputation

Density is recomputed every epoch. No historical density is permanently stored.

Remark 6. There is no reputation object. There is only sustained structural activity.

7 Income and Stasis

7.1 Purpose

Burn is thermodynamic sacrifice. Income is structural emergence.

Price and income are therefore decoupled:

- Burn differentiates signal from noise (thermodynamics).
- Income distributes metabolic capacity (cybernetics).

No event transfers price to another signer. Burn is destroyed. Income is minted from the epoch budget and allocated by structure.

7.2 Epoch Budget

Let $\mathcal{E}_T$ be the set of all valid events in epoch T .

Define epoch burn:

$$B_T = \sum_{e \in \mathcal{E}_T} b(e)$$

Let $\epsilon_T > -1$ be the macro-regulator state (Part V).

Define the minted epoch budget:

$$M_T = (1 + \epsilon_T) B_T$$

Thus the system remains thermodynamically anchored: more irreversible differentiation permits more realized distribution.

7.3 Baseline Scale

Define a robust baseline ripple for epoch T :

$$R_{\text{base}}(T) = \text{median}(\{R_T(e) \mid e \in \mathcal{E}_T\})$$

If $\mathcal{E}_T$ is empty, set $R_{\text{base}}(T) = 1$.

7.4 Stasis: Potential from Latency

Humans and creative systems require latency to organize internal structure. The protocol recognizes this as **potential**, not as immediate income.

For each signer a , define the last epoch in which a participated:

$$T_{\text{last}}(a)$$

Define inactivity duration:

$$\Delta T(a, T) = T - T_{\text{last}}(a)$$

Define stasis potential accumulator (bounded):

$$S_{\text{pot}}(a, T) = \min \left(S_{\text{max}}, S_{\text{pot}}(a, T-1) + s(\Delta T(a, T)) \right)$$

where $s(\cdot)$ is a nonnegative, increasing, concave function. Canonical choice:

$$s(\Delta) = \rho \cdot \ln(1 + \Delta)$$

with $\rho > 0$ and cap $S_{\text{max}} > 0$.

Remark 7. Stasis does not mint value by itself. It only becomes effective when validated by a subsequent structurally meaningful act.

7.5 Conditional Realization of Stasis

When signer a participates in an event $e \in \mathcal{E}_T$, their accumulated stasis may be realized only to the extent that the event creates real structure.

Define the stasis realized on event e :

$$S_{\text{real}}(a, e) = \min \left(S_{\text{pot}}(a, T), \eta_S \cdot DCD_T(e) \right)$$

where $\eta_S > 0$ is a protocol constant.

Update rule after the event (stasis is metabolically spent):

$$S_{\text{pot}}(a, T) \leftarrow S_{\text{pot}}(a, T) - S_{\text{real}}(a, e)$$

Interpretation:

- The Lazy: accumulates stasis, produces low density $\Rightarrow$ realization near zero.
- The Builder: accumulates stasis, produces high density $\Rightarrow$ system pays for silence.

7.6 Event Potential (Logarithmic Human Scaling)

Define a signer-level coefficient for event e :

$$C(a, e) = C_0 + S_{\text{real}}(a, e)$$

with baseline $C_0 > 0$.

Define event coefficient:

$$\bar{C}(e) = \frac{1}{|\text{Sig}(e)|} \sum_{a \in \text{Sig}(e)} C(a, e)$$

Define event potential weight:

$$P_T(e) = \bar{C}(e) \cdot \ln \left(1 + \frac{R_T(e)}{R_{\text{base}}(T)} \right)$$

This ensures:

- No winner-takes-all linear scaling (Weber–Fechner constraint),
- Structural causality is rewarded (via R_T),
- Rest matters only when validated (via S_{real}).

7.7 Income Allocation

If $\sum_{u \in \mathcal{E}_T} P_T(u) > 0$, define event income:

$$I_T(e) = M_T \cdot \frac{P_T(e)}{\sum_{u \in \mathcal{E}_T} P_T(u)}$$

Otherwise, set $I_T(e) = 0$.

Proposition 6 (Budget Identity).

$$\sum_{e \in \mathcal{E}_T} I_T(e) = M_T$$

7.8 Signer Income

By default, income is split equally among signers:

$$I_T(a) = \sum_{\substack{e \in \mathcal{E}_T \\ a \in \text{Sig}(e)}} \frac{I_T(e)}{|\text{Sig}(e)|}$$

Optional extension (teams): replace equal split with a deterministic share vector $\pi(a, e)$ with $\sum_{a \in \text{Sig}(e)} \pi(a, e) = 1$ and $\pi(a, e) \geq 0$.

7.9 Interpretation

- Burn is the thermodynamic cost of distinguishing an event from noise.
- Income is the cybernetic redistribution of capacity based on sustained structure.
- Stasis is metabolically necessary latency, recognized only when validated by impact.

Remark 8. You do not earn what others pay you. You earn what you enable.